17. The statistics behind Online Controlled Experiments

~~(4) Type I/I~~

(4) Type I/II Errors and Power I&J

1. Definitions / 1. Definitions.

- Type I Errors (α):
  Declaring a significant effect when none exists (false positive).
  ↳ E.g. you conclude that version B is better than A, but in reality there is no difference.

- Type II Errors (β):
  Failing to detect a real effect (false negative)
  ↳ E.g. you conclude no difference between A and B, but in fact B truly outperforms A.

- Statistical Power ($1-\beta$)
  The probability of correctly detecting a real effect (true positive rate)
  ↳ Power = 1 - Type II Error.

Real-world Analogy: Airport Security

Think of an airport metal detector screening passengers

- Null Hypothesis (H_0): Passenger is not carrying forbidden metal item.
- Alternative Hypothesis (H_1): Passenger is carrying forbidden metal item.

Decision by Detector	Reality: No Metal	Reality: Metal
Alarm (Flag)	Type I Error	Correct (Power)
No Alarm (Pass)	Correct	Type II Error.

- Type I (False Alarm): Innocent passenger flagged → inconvenience.
- Type II (Missed Detection): Dangerous item pass → security risk.

We must balance α and β: lowering α (fewer false alarms) often raises β (more misses), and vice versa.

(2) Mathematical Viewpoint.

Type I Error, Type II Error, and Power

For a one-sided hypothesis test in A/B testing, assuming normally data and known variance.

setup of A/B Test (Hypothesis Test for Difference in Means)

Workflow Testing

- Null Hypothesis (H_0): $\Delta = 0$ (no difference between A and B)
- Alternative hypothesis (H_1): $\Delta = \delta > 0$ (B is better than A by δ units)

Let:

- $\mu_A = \mu$
- $\mu_B = \mu + \delta$
- Population std dev: $\sigma = 10$
- Sample size per group: $n = 25$

So the sample means $\hat{x}_A$ and $\hat{x}_B$ are:

$$\hat{x}_A \sim N(\mu, \sigma^2/n),$$

$$\hat{x}_B \sim N(\mu + \delta, \sigma^2/n)$$

The difference in sample means:

$$\bar{D} = \bar{X}_{13} - \bar{X}_A \sim N(\delta, 2\sigma^2/n)$$

Step 1: Define Sampling Distributions:

Let's plug in values

- $\sigma = 10, n = 25, \delta = 5$

Then,

$$Var(\bar{D}) = \frac{2\sigma^2}{n} = \frac{2 \cdot 100}{25} = 8 \Rightarrow \bar{D} \sim N(0,8) \text{ under } H_0, N(5,8) \text{ under } H_1$$

Step 2: Choose Significance Level.

Let's use a one-sided test at $\alpha = 0.05$.

We now find the critical value c - the value of $\bar{D}$ above which we will reject H_0.

So under H_0: $\bar{D} \sim N(0,8)$, and we want:

$$P(\bar{D} > c \mid H_0) = 0.05$$

This is the 95th percentile of $N(0, \sqrt{8})$

We use the z-table:

- $z_{0.95} = 1.645$

Then,

$$c = \mu_0 + z_\alpha \cdot \sqrt{2\sigma^2/n} = 0 + 1.645 \cdot \sqrt{8} = 1.645 \times 2.828 \approx 4.65$$

So:

Decision Rule: Reject H_0 if $\bar{D} > 4.65$

Step 3: Compute Type I Error (α)

By design, we've already set:

$$\alpha = P(\bar{D} > 4.65 \mid H_0 : \bar{D} \sim N(0,8)) = 0.05$$

This is baked into how I picked the threshold.

Step 4: Compute Type II Error (β)

Now, under the alternative H_1, $\bar{D} \sim N(5, 8)$

we want the probability that we fail to reject H_0 even though H_1 is true:

$$\beta = P(\bar{D} \leq 4.65 \mid H_1 : \bar{D} \sim N(5, 8))$$

Standardized:

$$z = \frac{4.65 - 5}{\sqrt{8}} = \frac{-0.35}{2.828} \approx -0.124$$

Using the z-table:

$$P(Z \leq -0.124) \approx 0.4507 \Rightarrow \beta \approx 0.4507$$

Step 5: Compute Power ($1 - \beta$)

$$\text{Power} = 1 - \beta = 1 - 0.4507 \approx 0.5493.$$

Summary Table:

Metric	Value	Meaning
α	0.05	Probability of false positive (Type I)
Critical value	4.65	Threshold above which we reject H_0.
β	0.4507	Probability of a false negative (Type II).
power	0.5493	Chance of detecting true effect ($\delta = 5$).

④ Key Drivers of Power

- Effect Size (δ/σ): Larger true differences make detection easier (↑ power)
- Sample size (n): More data narrows sampling distributions (↓ β, ↑ power)
- Significance level (α): Higher α (looser threshold) raises power but ↑ Type I errors.

⑤ Overlap with other Topics

- Two-sample t-test (p. 186):

Relies on these error rates - its decision threshold is set by α; power determines its sensitivity.

- p-values & confidence intervals (p. 186):

A p-values below α triggers rejection of H_0 (Type I controlled), while CI width links to power (narrow CIs to detect small δ)

- Normality Assumption (p. 187):

Ensures the validity of α and β calculations for t-tests; CLT helps when n is large.

(5) ~~Pitfalls~~ Bias 191

1- What is Bias?

→ In the context of A/B testing, bias is a systematic error that makes your estimator consistently off-target formally, if you're estimating a parameter θ (e.g. the true lift Δ), and your estimator is $\hat{\theta}$, then:

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta.$$

- Zero bias ⇒ on average you hit the true value.
- Positive bias ⇒ you consistently overestimate.
- Negative bias ⇒ you consistently underestimate.

Analogy: A Miscalibrated Scale

Imagine a kitchen scale that always reads 2 kg too heavy

- True weight of flour bag = 1 kg
- The scale reports = 3kg

No matter how many times you weigh, you'll always be off by +2 kg. That constant 2 kg is the bias of that scale.

2. Bias in Experiments: Selection Bias.

In an A/B test, suppose:

↳ True conversion rates:

↳ A: 10%

↳ B: 15%

⇒ True lift δ = 5 ppt

↳ But you accidentally over-sample heavy users in B (they convert at 20%), while A is sampled correctly.

Then your observed B conversion might be ~20% making your estimated lift = 20 - 10 = 10 ppt, which is +5 ppt biased upward.

4) Math Behind Bias.

Let $\hat{\Delta}$ be your estimated lift from the experiment, and true lift is Δ. Then:

$$\text{Bias}(\hat{\Delta}) = \mathbb{E}[\hat{\Delta}] - \Delta$$

- If $\mathbb{E}[\hat{\Delta}] = \Delta$, estimator is unbiased.
- Otherwise, $\hat{\Delta}$ is biased.

5) Mitigating Bias.

- Run randomization: Ensures each user has equal chance in A or B.
- Stratification: Balance key segments (e.g. device type) before sampling.
- Instrumentation: Monitor for data loss or mis-tagging checks (cross-ref to "Instrumentation" in earlier chapters).
- Pre-experiment audits: Validate that no systematic differences exist in user attributes before rollout

4. Overlaps with Other Topics.

- Two-samples t-Test & p-values (pp 185-186):
  A biased sample may yield misleading p-values — you could claim significance when the true effect is smaller or zero.

- Type I / II Errors & Power (p. 186).
  Bias shifts your sampling distribution, inflating false positive (Type I) or masking real effects (increasing Type II)

- Multiple Testing (p. 191)
  If some variant. error systematically overestimated, due to partial data, the families wise error rate can sky rocket.

(6) Multiple testing (p. 192)

1. What is Multiple testing?

→ Multiple testing occurs when you perform many statistical tests at once. — For example, testing:

- many metrics (conversion, time on time, revenue...)
- many segments (new users, mobile users, U.S. users,)
- many variants (A/B/C/D...)

The more test you run, the more likely you are to find a false positive just by random chance.

Analogy

Imagine a metal detector (with 5% false alarm rate) scanning 20 people. Even if no one is carrying metal there a high chance the alarm will go off at least once.

⇒ In Experiments

You test 10 features on your website. For each:

- H_0: No impact
- H_1: There is impact.

You set $\alpha = 0.05$ for each. If all features have no true impact, you're still likely to see some "significant" results just due to chance. These are false discoveries.

2. Math Foundation

Concept 1: Family-wise Error Rate (FWER)

What is FWER?

→ If you're performing m independent hypothesis tests, each with significance level α, the family-wise Error Rate (FWER) is:

$$FWER = P(\text{At least one Type I error in } m \text{ tests})$$

Let's assume that all null hypothesis are true (global null hypothesis).

If tests are independent, the probability none of them result in a Type I error is:

$$P(\text{No Type I errors}) = (1-\alpha)^m$$

So:

$$FWER = 1 - (1-\alpha)^m$$

☑ Math Example

Let $m = 10$, $\alpha = 0.05$

$$FWER = 1 - (1-0.05)^{10} = 1 - (0.95)^{10}$$
$$\approx 0.4013$$

⇒ Even if all nulls are true, there's ~40% chance of making at least one false positive!

☑ Limiting Case:

As $m \to \infty$,

$$(1-\alpha)^m \to 0 \Rightarrow FWER \to 1$$

→ With enough test, you're guaranteed to get false positive unless you correct for it.

Concept 2: Bonferroni Correction

- Goal: Control FWER to be $\leq \alpha$.

Bonferroni's method simply adjusts the per-test significance level.

$$\boxed{\alpha^* = \frac{\alpha}{m}}$$

So each individual test use α^* and α.

Proof that it Controls FWER

Let V be the number of false positives (Type I errors).

Then:

By Boole's inequality (Union Bound):

$$P(V \geq 1) = P\left(\bigcup_{i=1}^{m} \text{Reject } H_{0i} \mid H_{0i} \text{ is true}\right) \leq$$

$$\sum_{i=1}^{m} P(\text{Reject } H_{0i}) = m \cdot \alpha^*$$

if $\alpha^* = \alpha/m$, then:

$$FWER \leq m \cdot \frac{\alpha}{m} = \alpha$$

✓ Bonferroni guarantees FWER $\leq \alpha$, regardless of dependence.

Example:

if $m = 20$, and you want FWER ≤ 0.05:

$$\alpha^* = \frac{0.05}{20} = 0.0025.$$

Every individual test must satisfy $p < 0.0025$ to be considered significant.

Downsides:

- Extremely conservative, especially for large m.
- Increases Type II errors (misses real effects).
- Better for few tests, independent tests.

Concept 3.: False Discovery Rate (FDR) - Benjamini-Hochberg.

Definition:

Let:

- V = Number of false positives.
- R = Number of rejections (significant tests)

Then FDR is:

$$FDR = \mathbb{E}\left[\frac{V}{R} \mid R > 0\right] \cdot P(R > 0)$$

(or just $\mathbb{E}\left[\frac{V}{R}\right]$)

☑ Benjamini - Hochberg (BH) Procedure:

Let $P_1, P_2, \ldots P_m$ be the p-values of your tests.

1. sort the p-values:

$$P_{(1)} \leq P_{(2)} \leq \ldots \leq P_{(m)}$$

2. Find largest k such that:

$$P_{(k)} \leq \frac{k}{m} \cdot \alpha$$

3. Reject all H_0's with ranks $i \leq k$

Example

Suppose $m = 5$ and sorted p-values are:

$$p = [0.002, 0.01, 0.04, 0.06, 0.09]$$

Want FDR controlled at $\alpha = 0.05$

check each:

i	p(i)	threshold = i/m × α
1	0.002	0.01
2	0.01	0.02
3	0.04	0.03 ✗ (fails)

☑ largest k = 2 such that $p_i \leq \frac{i}{m} \cdot \alpha$
→ Reject hypotheses 1 and 2

☑ Properties of BH
- less conservative than Bonferroni
- Control expected proportion of false positives.
- More powerful (less Type II errors)
- Works well when many hypotheses are true.

Comparison summary table

Metric	Formula	Controls	Conservative?	Power
FWER	$1-(1-\alpha)^m$	Probability of ≥1 Type I error	Very	Low
Bonferroni	$\alpha^* = \alpha/m$	FWER ≤ α	Yes	Low.
FDR (BH)	Rank-based threshold	Expected proportion of false discoveries	Moderate	High.

Examples

Bonferroni Correction.

Great! Let's now do a pure math example to show how Bonferroni correction mathematically controls the family-wise Error Rate (FWER) in multiple testing.

Goal: show with exact probability that Bonferroni correction keeps the total Type I error $\leq \alpha$.

Setup:

Let's test 3 null hypotheses:

- H_{01}, H_{02}, H_{03}
- All are truly null (no effect).
- We want to control FWER ≤ 0.05

⚠ Without Correction

Each test is run at $\alpha = 0.05$

Let A_i be the event "test i produces a false positive".

Then

$$P(A_i) = 0.05.$$

The probability of at least one Type I errors among the 3 tests is:

$$P(\text{at least one FP}) = P(A_1 \cup A_2 \cup A_3)$$

If the tests are independent then:

$$P(\text{no Type I error}) = (1 - 0.05)^3 = 0.95^3$$
$$= 0.857$$
$$\Rightarrow P(\geq 1 \text{ FP}) = 1 - 0.857 = 0.143$$

↑ FWER = 14.3% — clearly greater than 5% → unacceptable

☑ with Bonferroni Correction

we divide total $\alpha = 0.05$ among 3 tests:

$$\alpha^* = \frac{0.05}{3} \approx 0.0167$$

Now each test is conducted at level 0.0167

Then:

$$P(\text{no FP}) = (1 - 0.0167)^3 \approx 0.950$$

$$\Rightarrow \text{FWER} = 1 - 0.950 = 0.05$$

Insight:

- Bonferroni overestimates the true FWER unless tests are completely independent.
- But it is always safe. (controls the worst-case bound).
- This conservatism leads to more "lyze" errors (missed detections).

Benjamini-Hochberg FDR bound.

Goal: Given a set of p-values from multiple hypothesis tests, show how BH controls the FDR (False Discovery Rate), and which nulls we reject.

Setup: You conduct 5 hypothesis tests. The sorted p-values are:

$P_{(1)} = 0.004$, $P_{(2)} = 0.012$, $P_{(3)} = 0.030$, $P_{(4)} = 0.041$,

$P_{(5)} = 0.072$. You want to control FDR at $\alpha = 0.05$

Step-by-step: Benjamini-Hochberg Procedure.

We'll compute thresholds for each rank:

$$\text{Threshold} = \frac{i}{m} \cdot \alpha$$

Where:

- i = rank of the p-value
- m = total number of tests ($= 5$)

Rank i	$P(i)$	Threshold $\frac{i}{m} \cdot 0.05$	Compare	Reject?
1	0.004	0.01	$0.004 < 0.01$	☑ Yes
2	0.012	0.02	$0.012 < 0.02$	☑ Yes
3	0.030	0.03	$0.030 = 0.03$	☑ Yes
4	0.041	0.04	$0.041 > 0.04$	✗ No
5	0.072	0.05	$0.072 > 0.05$	✗ No.

Step 2 : Find the largest k such that:

$$P(k) \leq \frac{k}{m} \cdot \alpha$$

From table above:

- Largest $k = 3$
- So we reject all H_1, H_2, H_3

Result

- Rejection (R) = 3
- If we know from the data that only 2 of these 3 were truely null, then:
  - False Dis coveries $V = 1$
  - $FDR \simeq \frac{V}{R} = \frac{1}{3} \simeq 0.3333.$

→ But during many tests and repetitions, the expected value of this proportion stays $\leq \tau =$

0.05 (under certain assumptions, like independence).

⇒ Math Definition Reminder

$$FDR = E\left[\frac{V}{R}\right], \quad V = \text{False Positives}, \quad R = \text{Total Rejections.}$$

The BH procedure controls this expected ratio.

⇒ Summary

Step	Action
Sort p-values	Ascending
Compute BH thresholds	$\frac{i}{m} \cdot \alpha$
Find largest k	where $P(k) \leq$ threshold.
Reject H_1 to H_k	These hypotheses are called significant.

③ Overlaps with Other Topics.

Topic	Relationship
Type I Error (p. 189)	→ Multiple tests increase overall Type I error rate.
Bias (p. 191)	→ If you only report significant results from many tests, this selection bias.
p-values (p. 186)	→ p-values must be corrected or interpreted in context of multiple tests.
Fisher's Meta-analysis (p. 192)	→ A method to ~~continue~~ combine multiple p-values correctly.

(4) Summary Table

Term	Meaning
Multiple Testing	→ Running many statistical tests on same or related data.
FWER	→ Probability of ≥ 1 Type I Type I across all tests
Bonferroni Correction	→ Adjust α per test: α/m
FDR (Benjamini-Hochberg)	→ Controls expected proportion of False positives.

(7) Fisher's Meta-analysis.

1. What is Fisher's Meta-analysis?

→ Fisher's method is a way to combine p-values from multiple independent test of the same hypothesis into a single test statistic. It answers:

> "Given several experiments (or segments), is there overall evidence against the null?"

If you have p-values $p_1, p_2, \ldots p_k$, Fisher's combined statistic is:

$$X^2 = -2 \sum_{i=1}^{k} \ln(p_i),$$

Under the global null (all tests true null), X^2 follows a chi-square distribution with $2k$ degrees of freedom.

Why and when to use it

- Aggregation: when you run the same A/B test in multiple independent segments (e.g. countries, devices) and want an overall conclusion.
- Consistency: If each segment's p-values is only moderately significant, combining them can reveal a stronger joint signal.

Real-world Analogy: Jury Deliberation.

Imagine a jury of k people each casting a "not-guilty" or "guilty" vote. Each juror's confidence in their "guilty" vote is represented by a p-value:

- Small p-value ⇒ high confidence of guilt
- Large p-value ⇒ low confidence.

Fisher's method aggregates these confidences into one statistic that the judge uses to decide guilt overall.

2. Step-by-step pure math example.

Goal: Combine several p-values from different tests into a single p-value that tells us if there's an overall significant effect.

Step-by-step Math Example:

You run 3 independent A/B test on different platforms:

- Web: p-value = 0.04
- Mobile: p-value = 0.06.
- Tablet: p-value = 0.01.

Step 1: Use Fisher's formula

$$X^2 = -2 \sum_{i=1}^{k} \ln(p_i)$$

$$X^2 = -2 [\ln(0.04) + \ln(0.06) + \ln(0.01)]$$

$$= -2 [-3.2189 - 2.8134 - 4.6052]$$

$$= -2 \times (-10.6373) = 21.275.$$

Step 2: Determine Degree of Freedom

$$df = 2k = 2 \times 3 = 6.$$

Step 3: Use the chi-square Distribution

Under the null hypothesis (all three tests are truely null), this X^2 follows a chi-square distribution with 6 degree of freedom.

To compute the combined p-value:

$$P(\chi^2_6 \geq 21.275)$$

output

$x^2 = 21.275$

Combined p-value $= 0.0016$.

☑ Interpretation

- Even though none of the p-values were below 0.01 individually.
- When combined, they show strong evidence against the global null.
- So, Fisher's method amplifies weak but consistent signals.

Summary Table

p-values	$\ln(P_i)$
0.04	-3.2189
0.06	-2.8134
0.01	-4.6052
Sum	-10.6375

$-2 \sum \ln P_i$ 21.275.

$df = 2k = 6$

Final p-value $= 0.0016$ ☑ Significant.

3. What Problem Does it Solve?

Suppose you run the same experiment on:

- Different countries
- Different devices.
- Different user segments

Each segment gives you a p-value, You want to answer:

> "Is there overall evidence that the treatment had a effect across these segments?"

This is where Fisher's method helps. It tests the global null hypothesis:

> All local null hypotheses are true (i.e. no effect in any segment).

4. Core formula

If you have k independent p-values $p_1, p_2, \ldots, p_k$ Fisher's test statistic is:

$$\chi^2 = -2 \sum_{i=1}^{k} \ln(p_i)$$

This χ^2 statistic follows a chi-square distribution with 2k degrees of freedom under the global null hypothesis.

Then we calculate:

$$\text{Combined p-value} = P(\chi^2_{2k} \geq \chi^2)$$

Chi-Square Test Overview

(1) Intro

- Purpose: Tests whether observed data counts differ significantly from expected counts.
- Most common use: Contingency tables - to assess if variables are independent (e.g., if click rate is independent of headline type).
- Origin: Introduced by Karl Pearson (1900)
- Name origin: "Chi" (X^2) is the Greek letter used for the test statistic.

(2) Key Terms

Chi-square statistic (X^2)

Sum of squared differences between observed and expected values, normalized by the expected value.

$$X^2 = \sum \frac{(O - E)^2}{E}$$

Expected counts (E):

What we'd expect under the null hypothesis, often assuming uniform or proportional distributions.

Degrees of freedom (df)

$$df = (r - 1) \times (c - 1)$$

where r = number of rows,

c = number of columns in contingency table.

Web A/B/C Testing Example

→ Testing three headline (A, B, C) each on 1,000 users.

→ Observed clicks
- A: 14
- B: 8
- C: 12
- Total clicks = 34 → Overall click rate = 34/3000 = 1.13%

→ Expected under null hypothesis (same click rate)
- Clicks for each = 11.33
- No. clicks = 988.67

→ Pearson Residuals $(O-E)/\sqrt{E}$
- Measure how much each cell deviates from expected value

→ Chi-square statistics
- sum of squared residuals = 1.666.

Example 1: chi-Square Test for a Fair Die (Goodness-of-fit test)

~~? Question: Is a die fair?~~

? Question: Is a die fair?

You roll a 6-sided die 60 times. You observe:

Face	Observed (O)
1	8
2	10
3	9
4	11
5	12
6	10

If the die is fair, each face should appear 10 times (Expected, E).

Step 1: Compute Chi-Square statistics.

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

$$\chi^2 = \frac{(8-10)^2}{10} + \frac{(10-10)^2}{10} + \frac{(9-10)^2}{10} + \frac{(11-10)^2}{10} + \frac{(12-10)^2}{10} + \frac{(10-10)^2}{10}$$

$$\chi^2 = 4/10 + 0 + 1/10 + 1/10 + 4/10 + 0 = \frac{10}{10} = 1.0$$

Step 2: Degrees of freedom

$$df = k - 1 = 6 - 1 = 5$$

Step 3: Compare with critical value or compute p-value.

The critical value is the cutoff point on the chi-square distribution beyond which we consider the result statistically significant (i.e. we reject the null hypothesis).

It depends on:

- Degree of freedom (df)
- Significance level (α) — commonly 0.05, 0.01 or 0.10.

Use a chi-square table

Critical χ^2 at $df = 5$, $\alpha = 0.05 \rightarrow$ ~~3~~ 11.07

Since $10 < 11.07$, we fail to reject the null hypothesis

✓ Conclusion: The die appears fair.

Example 2 : Chi-square Test of Independence

? Question : Are gender and handedness independent?

You survey 100 student

	Left-handed	Right-handed	Total
Male	10	40	50
Female	5	45	50
Total	15	85	100

Step 1 : Compute Expected Counts

$$E_{row,\,col} = \frac{(\text{row total} \times \text{col total})}{\text{grand total}}$$

E.g. For Male - Left

$$E = \frac{50 \times 15}{100} = 7.5$$

All expected counts

	Left (E)	Right
Male	7.5	42.5
Female	7.5	42.5

Step 2 : Compute Chi-Square Statistics.

$$\chi^2 = \sum \frac{(O-E)^2}{E} = \frac{(10-7.5)^2}{7.5} + \frac{(40-42.5)^2}{42.5} + \frac{(5-7.5)^2}{7.5} + \frac{(45-42.5)^2}{42.5}$$

$$\chi^2 = \frac{6.25}{7.5} + \frac{6.25}{42.5} + \frac{6.25}{7.5} + \frac{6.25}{42.5}$$

$$= 0.833 + 0.147 + 0.833 + 0.147 = 1.96.$$

Step 3: Degree of freedom

$$df = (2-1)(2-1) = 1$$

Critical χ^2 at $df = 1$, $\alpha = 0.05 \rightarrow 3.841$

Since $1.96 < 3.841$, we fail to reject independence.

☑ Conclusion: No significant relationship between gender and handedness.

(3) Resampling-based chi-square test.

Problem setup

You have a set of 45 total observations.

- 9 are **"1"**s (representing "success" or "clicks")
- 36 are 0's (representing "no success")

You randomly divide these 45 items into 3 groups of 15 each.

? Goal

Do the groups differ more than expected by chance in how many "1"s they contain?

Step 1: Observed data

Suppose after one real-world observation, you get:

Group	Observed 1s	Observed 0s
A	4	11
B	2	13
C	3	12
Total	9	36

Step2: Expected Counts under Null Hypothesis.

Assuming all groups should have equal success rate, the overall success rate is: 20%.

Each group has 15 trials, so expected successes per group:

$$E = 15 \times 0.2 = 3$$

Expected failures:

$$15 \times 0.8 = 12.$$

Group	Expected 1s	Expected 0s.
A	3	12
B	3	12
C	3	12

Step3: Compute Observed χ^2 statistics

Use:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

We do this across all 6 cells (1s and 0s for each group):

$$\chi^2 = \frac{(4-3)^2}{3} + \frac{(11-12)^2}{12} + \frac{(2-3)^2}{3} + \frac{(13-12)^2}{12} + \frac{(3-3)^2}{3} + \frac{(12-12)^2}{12}$$

$$= \frac{1}{3} + \frac{1}{12} + \frac{1}{3} + \frac{1}{12} + 0 + 0$$

$$= 2\left(\frac{1}{3} + \frac{1}{12}\right) = \frac{2}{3} + \frac{1}{6} = \frac{5}{6} \approx 0.833$$

$\Rightarrow$ Observed $\chi^2 = 0.833$.

Step4 : Simulate Resampling (Permutation Test)

Need simulate one permutation example.

↳ Take the 9 ones and 36 zeros
↳ Shuffle randomly.
↳ Assign first 15 to Group A, next 15 to Group B, final 15 to Group C.
↳ Count how many 1s in each group.
↳ Compute new χ^2 using same formulas.

Repeat this many times (e.g. 1000 simulations). For each simulation:

1. Count how many 1s per group.
2. Use expected = 3.
3. Compute chi-square.
4. save the values.

Step5 : Compute p-value

$$\text{p-value} = \frac{\#(\text{simulated } \chi^2 \geq \text{observed } \chi^2)}{\text{total simulations.}}$$

E.g. If 374 out of 1000 simulated chi-square values were ≥ 0.633

$$p = \frac{374}{1000} = 0.374$$

If this p-value is large, the variation you saw is consistent with chance.

Final thoughts (Pure Math Version)

This example does not require any application domain — it's just:

- Binary data (0s and 1s)
- Divided into equal groups
- compared against equal-probability null hypothesis.
- Tested via resampling to estimate how likely such a distribution is under randomness.

(1) Theoretical Chi-Square Test.

Problem

Suppose we have this 2×3 contingency table from a theoretical experiment:

Group	Category A	Category B	Category C	Row Total
X	10	20	30	60
Y	20	10	10	40
Column Total	30	30	40	100

We want to test independence between rows and columns using the theoretical chi-square test.

Step 1: Compute Expected Counts

$$E_{ij} = \frac{(\text{row total}) \times (\text{column total})}{\text{grand total}}$$

Compute expected values for each cell:

- Cell (X, A):

$$\sum X_A = \frac{60 \times 30}{100} = 18$$

- Cell (X, B):

$$E_{XB} = \frac{60 \times 30}{100} = 18$$

- Cell (X, C)

$$E_{XC} = \frac{60 \times 40}{100} = 24$$

- cell (Y, A)

$$E_{YA} = \frac{40 \times 30}{100} = 12$$

- cell (4, B)

$$E_{4B} = \frac{40 \times 30}{100} = 12$$

- cell (4, C)

$$E_{4C} = \frac{40 \times 40}{100} = 16.$$

so, expected table:

Group	A(E)	B(E)	C(E)
X	18	18	24
4	12	12	16

Step 2: Compute X^2 statistic

Formula:

$$X^2 = \sum \frac{(O-E)^2}{E}$$

let's compute each term

Cell	O	E	$(O-E)^2/E$
X, A	10	18	$(-8)^2/18 = 3.56$
X, B	20	18	$(12)^2/18 = 4/18 = 0.22$
X, C	30	24	$(6)^2/24 = 1.5$
4, A	20	12	$(8)^2/12 = 5.33$
4, B	10	12	$(-2)^2/12 = 0.33$
4, C	10	16	$(-6)^2/16 = 2.25$

Add them up:

$$X^2 = 3.56 + 0.22 + 1.3 + 5.33 + 0.33 + 2.25 = 13.09$$

Step 3: Degrees of freedom

$$df = (\text{rows} - 1)(\text{columns} - 1) = (2-1)(3-1) = 2$$

Step 4: Compare to chi-square.

We use a chi-square distribution with $df = 2$ to compute the p-value. You can use a chi-square table or calculator

This gives: $p \approx 0.00137$.

Final Conclusion

Since:

- $\chi^2 \approx 13.19$
- $df = 2$
- $p \approx 0.0014 < 0.05$

Reject the null hypothesis: there is a statistically significant relationship between groups and category.

Summary

Step	Value
Observed χ^2	13.19
Degree of freedom	2
Critical χ^2 ($\alpha = 0.05$)	5.991
p-value	0.00137
Decision	Reject H_0.

(5) Fisher's Exact Test.

Purpose

Fisher's Exact Test tests whether two categorical variables are independent (like the chi-square test), but is used especially when the sample size is small, and cell counts are low (especially < 5).

Key ideas:

- Based on exact enumeration of all possible 2×2 tables that preserve the row and column totals
- Calculates exact p-value: the probability of seeing the observed table or a more extreme one, under the null hypothesis (independence).
- Uses the hypergeometric distribution to compute probabilities.

Example: Fisher's Exact Test (Pure Math)

Problem

Suppose we have this 2×2 contingency table:

	Success	Failure	Row Total
Group A	1	4	5
Group B	4	0	4
Col Total	5	4	9

We want to test:

Is success independent of group?

Step 1 : Define the Fixed Margins.

- Total "success" = 5
- Total "failure" = 4
- Row totals :
  - ↳ Group A has 5 total
  - ↳ Group B has 4 total

* These total are fixed*, and under the null hypothesis all valid 2×2 tables with these margins are equally possible

Step 2 : Enumerate All Possible Tables.

* Only the top-left cell (A - success) determines the full table. Let's denote this value as x*. It can range from 0 to 5 (min (total successes, GroupA size)).

For each value of $x = 0, 1, \dots, 5$, we construct the full table and compute its probability using the hypergeometric formula.

$$P(x) = \frac{\binom{a}{x}\binom{b}{r-x}}{\binom{a+b}{r}}$$

where :

- a = number in row A = 5
- b = number in row B = 4
- r = total number of successes = 5
- x = number of successes in Group A

Compute All Probabilities.

x (A-Success)	A Table	Probability (p)
0	[0,5], [5,0]	$\frac{\binom{5}{0}\binom{4}{5}}{\binom{9}{5}} = 0$ (since $\binom{4}{5}=0$)
1 (observed)	[1,4], [4,0]	$\frac{\binom{5}{1}\binom{4}{4}}{\binom{9}{5}} = \frac{5\times1}{126} = \frac{5}{126} \simeq 0.0397$
2	[2,3], [3,1]	$\frac{\binom{5}{2}\binom{4}{3}}{\binom{9}{5}} = \frac{10\times4}{126} = \frac{40}{126} \simeq 0.3175$
3	[3,2], [2,2]	$\frac{\binom{5}{3}\binom{4}{2}}{\binom{9}{5}} = \frac{10\times6}{126} = \frac{60}{126} = 0.4762$
4	[4,1], [1,3]	$\frac{\binom{5}{4}\binom{4}{1}}{\binom{9}{5}} = \frac{5\times4}{126} = \frac{20}{126} \simeq 0.1587$
5	[5,0], [0,4]	$\frac{\binom{5}{5}\binom{4}{0}}{\binom{9}{5}} = \frac{1\times1}{126} = \frac{1}{126} = 0.0079$

Step3 : Calculate p-value

You sum the probabilities of all tables that are as extreme or more extreme than the observed.

- Ours observed value : $x = 1$ (Group A has 1 success)
- Under a two-sided test, we includes all tables whose probabilities are less than or equal to that of the observed one (convention ~~of~~ for Fisher's test in most softwares)

Quick Revision

** Only the top-left cell (A - Success) determines the full tables **

#1. You start with a 2×2 table like this:

	Success	Failures	Row Total
Group A	?	?	5
Group B	?	?	4
Col Total	5	4	9

These totals are fixed. That's important:

- 5 people are in Group A.
- 4 people are in Group B
- 5 total "successes"
- 4 total "failures".

#2. Now fix only one cell — the top-left cell, which is Group A, successes — let's call this value x

Let's suppose:

| x = 1 (i.e. 1 success in Group A)

Now watch how the entire table can be filled in automatically:

	Success	Failure	Row Total
Group A	$x = 1$	$5 - x = 4$	5
Group B	$5 - x = 4$	$4 - (5 - x) = 0$	4
Total	5	4	9

Let's verify the math:

- If Group A has 1 success → Group A has 4 failures (because total in Group A = 5)
- Total successes = 5, so Group B must have $5 - 1 = 4$ successes.
- Total failures = 4, so Group B must have 0 failures.

✓ Table is complete - and you only needed $\boxed{x = 1}$ to build it.

#3. Why does this work?

Because:

- Row totals and column totals are fixed (pre-determined)
- So when you know one cell, the others are just subtraction.

This is what makes Fisher's Exact Test a fixed-margins test:

> one cell (usually top-left) uniquely determines the rest 2×2 table.

** Observed Table. **

The only cell that varies is Group A, success, which we'll call $\boxed{x}$
we'll compute probabilities of each valid table ($x = 0$ to 5) using:

Hypergeometric formula

$$P(x) = \frac{\binom{5}{x} \cdot \binom{4}{5-x}}{\binom{9}{5}}$$

- Top: ways to choose $\boxed{x}$ successes in Group A (out of 5 people).
- Then: $\boxed{5-x}$ success in Group B (4 people)
- Bottom: all ways to choose 5 successes from 9 people total.

we calculate

- $\binom{9}{5} = 126$ → fixed denominator for all

$\boxed{\checkmark}$ $x = 0$

- Group A has 0 successes → Group B must have all 5.
- But Group B only has 4 people → not possible

$$P(0) = \frac{\binom{5}{0} \cdot \binom{4}{5}}{126} = \frac{1 \cdot 0}{126} = 0.$$

$\boxed{\checkmark}$ $x = 1$ (observed table)

	Success	Failure
Group A	1	4
Group B	4	0

$$P(1) = \frac{\binom{5}{1} \cdot \binom{4}{4}}{126} = \frac{5 \cdot 1}{126} = \frac{5}{126} \approx 0.0397$$

☑ $x = 2$

	Success	Failure
Group A	2	3
Group B	3	1

$$P(2) = \frac{\binom{5}{2} \cdot \binom{4}{3}}{126} = \frac{10 \cdot 4}{126} = \frac{40}{126} \approx 0.3175.$$

☑ $x = 3$

	Success	Failure.
Group A	3	2
Group B	2	2

$$P(3) = \frac{\binom{5}{3} \cdot \binom{4}{2}}{126} = \frac{10 \cdot 6}{126} = \frac{60}{126} \approx 0.4762$$

☑ $x = 4$

	Success	Failure.
Group A	4	1
Group B	1	0

$$P(4) = \frac{\binom{5}{4} \cdot \binom{4}{1}}{126} = \frac{5 \cdot 4}{126} = \frac{20}{126} = 0.1587$$

☑ $x = 5$

	Success	Failure.
Group A	5	0
Group B	0	4

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$$p(5) = \frac{\binom{5}{5} \cdot \binom{4}{0}}{126} = \frac{1 \cdot 1}{126} = \frac{1}{126} \approx 0.0079.$$

Continue step 3 ----

How to compute the p-value in Fisher's Exact Test.

1. In Fisher's test (two-tailed), the p-value is the sum of probabilities of all tables that are at least as extreme as the observed table.

But what does "extreme" mean?

- It's not based on how far x is from expected, but on how unusual the table is under the null.
- That's done by comparing the probabilities of all possible tables.

Observed value:

- $x = 1$, $P(x=1) = 0.0397$

Now the key logic for Fisher's two-tailed p-value:

☑ Add probability of all tables whose probability is $\leq$ the observed one (This is the convention used by Fisher.test() in and most softwares).

Tables with probability ≤ 0.0397

x	P(x)	
1	0.0397	☑ (observed)
5	0.0079	☑ (even more extreme)
0	0	☑ (invalid but ≤ 0.0397)

so we add!

p-value = P(x=1) + P(x=5) + P(x=0)
= 0.0397 + 0.0079 + 0 = 0.0476

Final Answer

- Fisher's Exact Test p-value = 0.0476
- Since this is < 0.05, we reject the null hypothesis at 5% level.

Summary of Fisher's Logic

Term	Meaning
Observed Table	x=1, Group A had 1 success
Observed Probability	P(x=1) = 0.0397
Rule for Two-tailed p	Add probabilities of tables with P ≤ 0.0397
Added Tables	x=0 (0), x=1 (0.0397), x=5 (0.0079)
Final p-value	0.0476.

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7. Real-world Case: Scientific Fraud Detection.

→ Investigated abnormal frequency of digits in lab data (Imanishi-kari case)
→ Expected uniform distribution of interior digits (0-9)
→ used chi-square test to detect non-random pattern.
→ Example of Benford's Law-style application.

8 Key Insights & Use cases

Scenario	Suggested Method.
Large sample size, expected counts ≥ 5	→ Chi-Square test (chisq.test(x))
Small sample size, expected counts < 5	→ Fisher's Exact Test (fisher.test(x))
Need for accuracy / custome behavior modeling	→ Resampling / Permutation Test. (chisq.test(x, simulate.p.value=TRUE))
Optimal choice selection	→ Multi-Armed Bandits
Feature selection for categorical variables	→ X^2 feature selection in ML.